

Statement

- The accepted gold standard for burn first aid is **20 minutes of cool clean running water**
- If clean *running* water is not available; immersion in cool water or application of wet towels (2 towels alternated every 15 seconds) may have limited usefulness for first aid
- Evidence indicates that the cooling function of hydrogel dressing products is not as effective as cool running water for first aid
- If no water is available, application of hydrogel or other wet dressings may be useful as an analgesic but should be replaced by water as soon as available if within 3 hours of injury
- The patient must be kept warm during cooling of the burn wound – **‘Cool the Burn: Warm the Patient’**
- After first aid, patients with large burns should be covered in dressings that minimise evaporative heat loss by excluding air from the wound and the outer surface of the dressing.
- For transfer to hospital, plastic cling films are suitable if less than 8 hours or if dressings are not available.

First Aid for Burns

- The accepted gold standard for burn first aid is 20 minutes of cool running water applied to the burn as soon as possible (within 1 hour is best but up to 3 hours has effect) after injury
- This is especially applicable to dermal thickness wounds, where cooling may decrease burn wound progression and improve healing

Hydrogel dressings

- Hydrogel dressings have been advertised for the acute management of burn injuries
- Some are promoted as providing cooling of the wound and a dressing function
- Moist dressings rely on exposure to air for their cooling effect via evaporative heat loss
- This may be helpful for small burns (<10% Total Body Surface Area [TBSA]) in the first few hours after injury, but in patients with more extensive injury, can contribute to hypothermia

Analgesia

- Cool moist dressings such as hydrogels may provide good analgesia for dermal thickness burns after appropriate first aid and before definitive dressings are applied

Hypothermia

- Patients with extensive burn wounds (>20%TBSA in adults or >10%TBSA in children) are at increased risk of hypothermia - especially in children and the elderly.
- Hydrogels, or any wet dressing, can also be associated with development of hypothermia if exposed to the air and left in place for prolonged periods, particularly in the elderly or children with larger burns, and should be avoided
- Patients should be monitored closely for hypothermia. Keep warm during cooling using heated blankets and warmed environments.
- Cooling should be discontinued if hypothermia develops